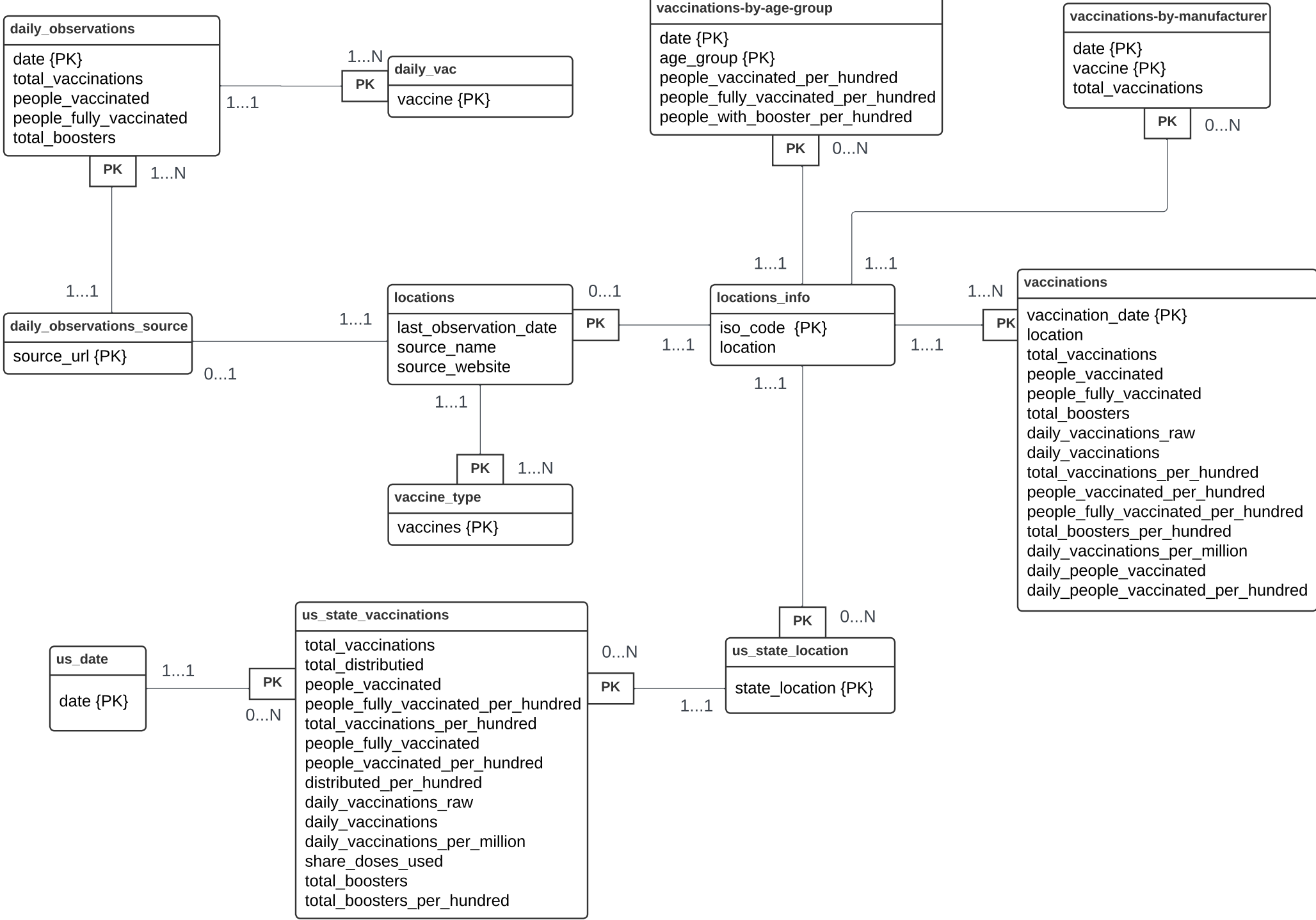


ER DIAGRAM



Explanation of normalisation challenges and the resulting changes :-

- Created a separate table '**locations_info**' to store the data of country, continent, union, etc., names and their respective iso codes. Then removed the redundant column 'location' from the table '**locations**'. Also, removed comma separated list of the attribute 'vaccines' from the table '**locations**' keeping a separate table '**vaccine_type**' to store the types of vaccines. Fixed a repeated data value 'Pfizer/BioNTech' from the same list in the row 16. Brought the table to 3NF.
- Split the table '**us_state_vaccinations**' into three tables '**us_date**', '**us_state_vaccinations**' and '**us_state_location**' which resulted in increasing consistency of the dataset for a database design.
- Removed the redundant attribute 'location' from the table '**vaccinations**' as the data can be retrieved from the table '**locations_info**'. Brought it to 3NF.
- Combined the data from the tables '**Australia**', '**United States**', '**England**' and '**New Zealand**' into a single table named '**daily_observations**' for consistency of the dataset.
- Stored the information of source url and source identity in a table '**daily_observations_source**'. Removed the redundant attribute named 'location' from the table '**daily_observations**' as the information from its data can be retrieved from the table '**daily_observations_source**'. Brought it to 3NF.
- Replaced an attribute named 'location' from the tables '**vaccinations-by-manufacture**' and '**vaccinations-by-age-group**' with 'iso_code', replaced the country names with their respective iso codes. Brought them to 3NF.
- Removed comma separated list of the attribute 'vaccine' from the table '**daily_observations**', keeping a separate table '**daily_vac**' to store the types of vaccines related to the table '**daily_observations**'. Brought the table to 3NF.

Database Schema :-

- locations_info** (iso_code, location)
- us_date** (date)
- daily_observations_source** (source_url , observation_iso*)
- locations** (iso_code*, last_observation_date, source_name, source_website)
- us_state_location** (state_location)
- vaccinations** (all iso_code*, vaccination_date, people_vaccinated, people_fully_vaccinated, total_boosters, daily_vaccinations_raw, daily_vaccinations, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, total_boosters_per_hundred, daily_vaccinations_per_million, daily_people_vaccinated, daily_people_vaccinated_per_hundred)
- vaccinations-by-manufacturer** (iso_code*, date, vaccine, total_vaccinations)
- vaccinations-by-age-group** (iso_code*, date, age_group, people_vaccinated_per_hundred, people_fully_vaccinated_per_hundred, people_with_booster_per_hundred)
- us_state_vaccinations** (date*, state_location*, total_vaccinations, total_distributed, people_vaccinated, people_fully_vaccinated_per_hundred, total_vaccinations_per_hundred, people_fully_vaccinated, people_vaccinated_per_hundred, distributed_per_hundred, daily_vaccinations_raw, daily_vaccinations, daily_vaccinations_per_million, share_doses_used, total_boosters, total_boosters_per_hundred)
- daily_observations** (source_url*, date, total_vaccinations, people_vaccinated, people_fully_vaccinated, total_boosters)
- vaccine_type** (vac_iso_code*, vaccines)
- daily_vac** (source_url*, date *, vaccine)